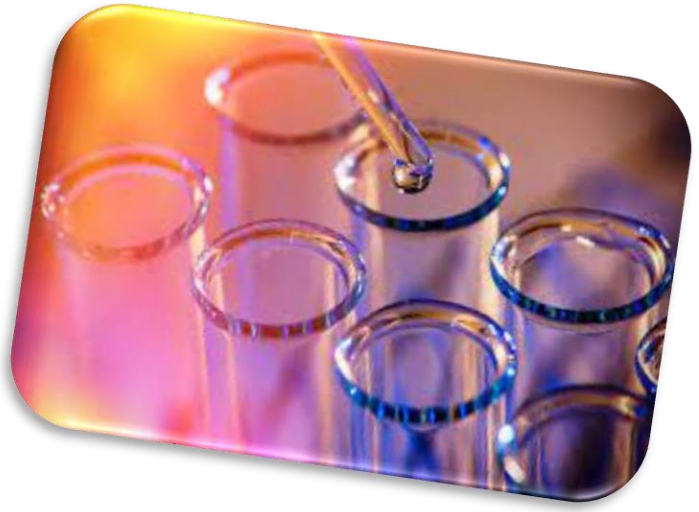
Arab Academy
for Science, Technology & Maritime Transport

Biochemistry
EBA1300





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Lecture 8

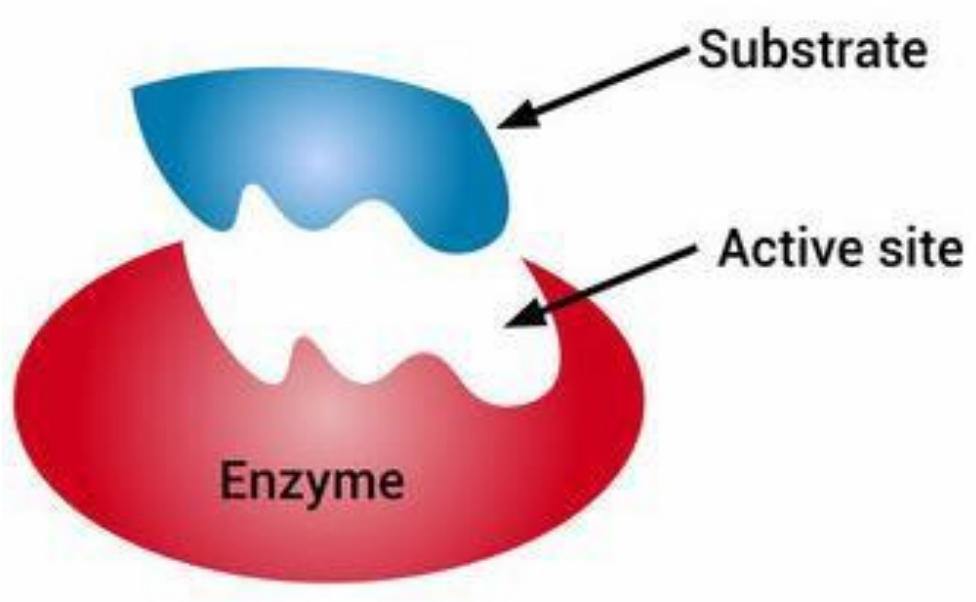
Enzymes (part 1)

Important to know

Enzyme

Active site

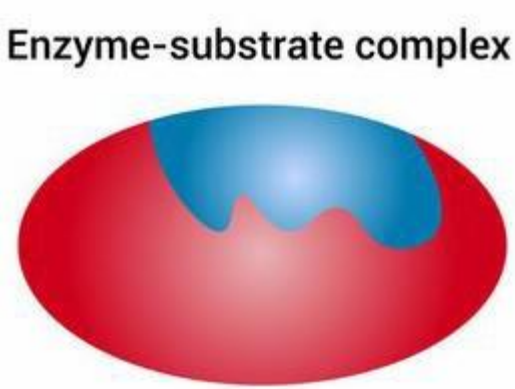
Substrate

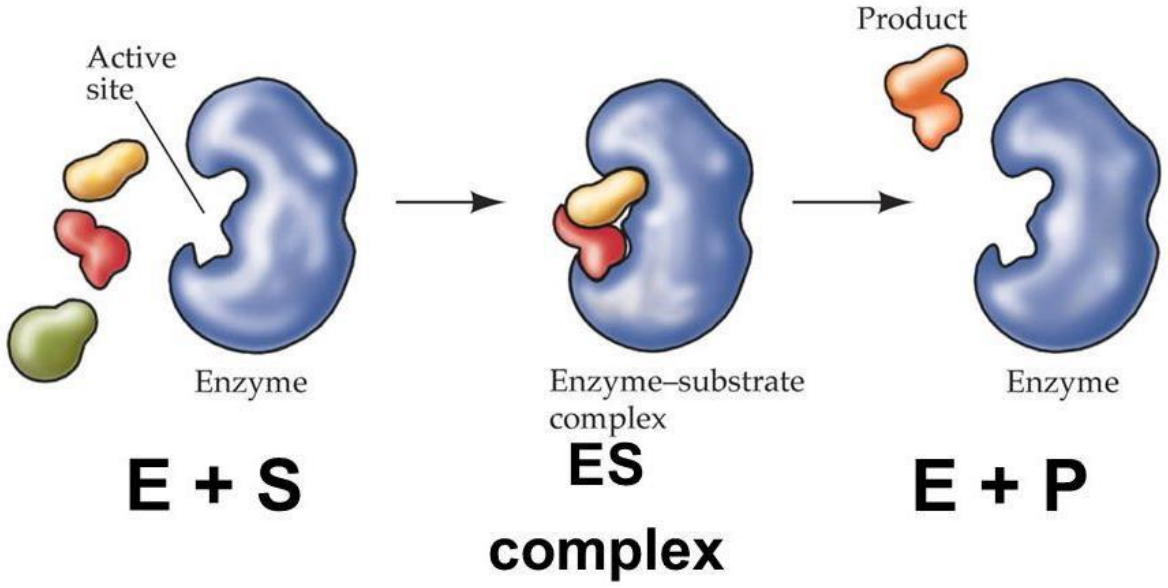


Important to know

ES complex

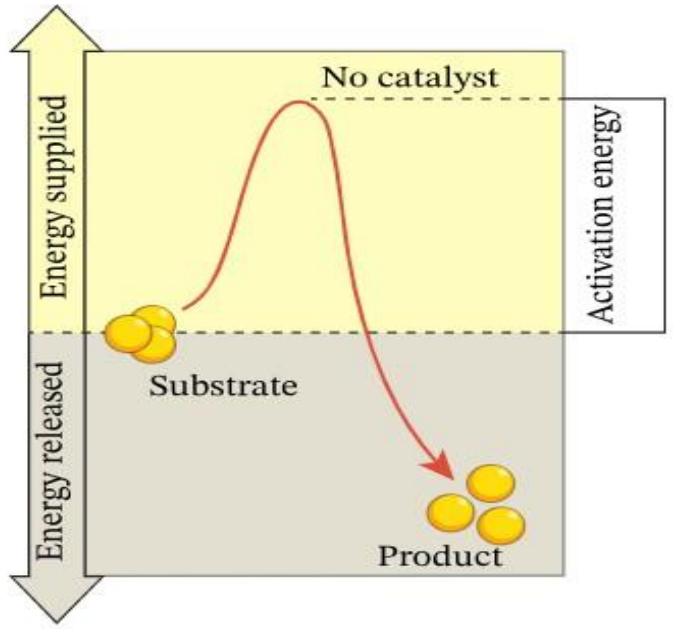
Product



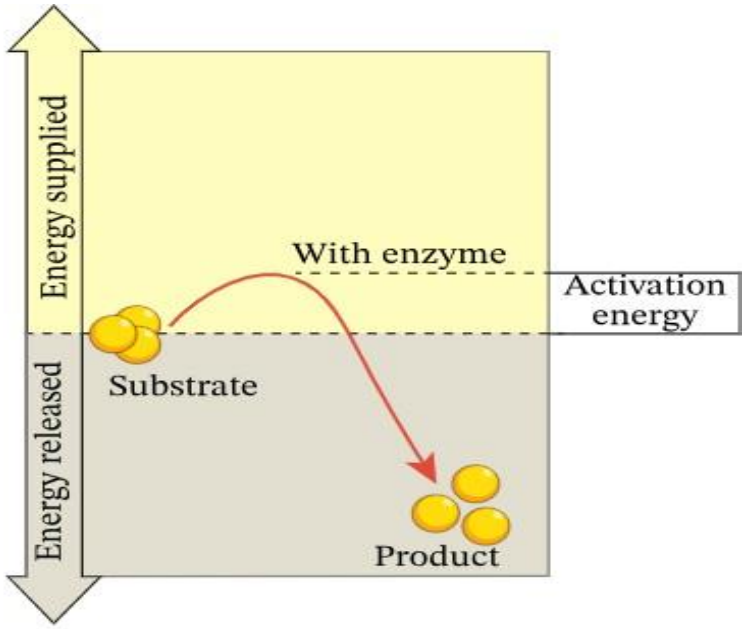


Important to know

Enzyme Catalysis → **Increase Rate of reaction** → **Decrease Activation energy**



Progress of reaction without enzyme



Progress of reaction with enzyme

Enzymes

- Enzymes are important for all reactions in the body.
- **What are enzymes?**
  - Enzymes are protein catalysts that increase the rate of reactions (accelerate reaction) without being changed.
- They increase the velocity of a biochemical reaction, and are not consumed during the reaction.
- Molecules that react with enzymes are called **substrates**.

Enzymes

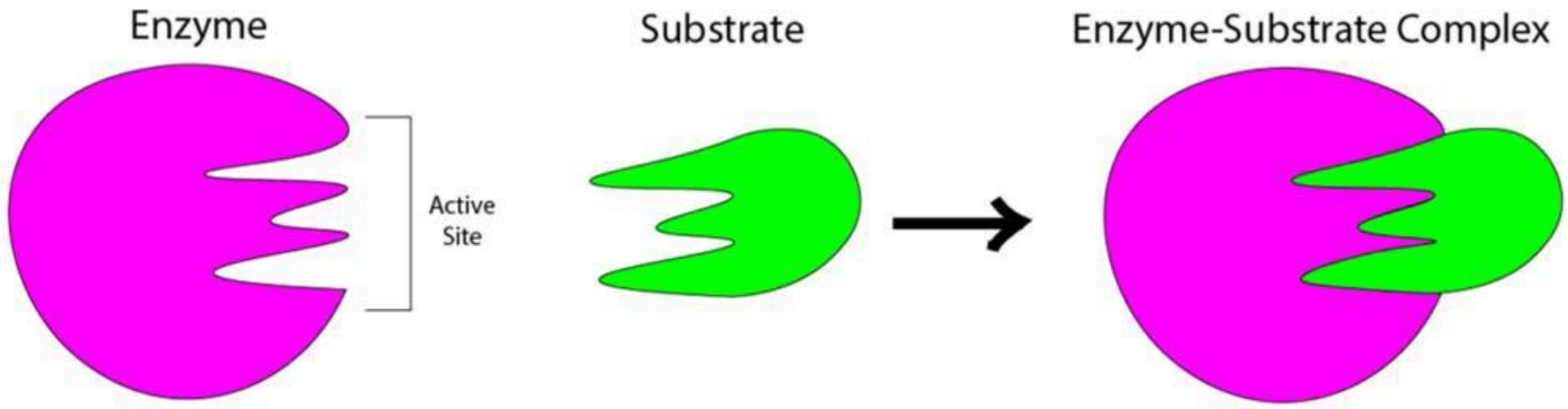
- Most commonly used enzyme names have the suffix "-ase"
  - Ex. lipase , lactase , dehydrogenase.

- Some enzymes retain their original names, without "-ase" , for example, trypsin and pepsin.

- The material on which the enzyme will act is called the substrate.

Enzymes

- The substrate binds to enzyme at a specific location called the **active site**.
- When the enzyme binds to the substrate, they give **enzyme-substrate complex**.



General properties of enzymes

1 Enzymes are proteins.

2 Enzymes accelerate the reaction but:

a. Are Not consumed in the reaction.

b. Are required in very small quantities.

3 Enzymes are **highly specific** for their substrate
(each substrate reacts with its enzyme).

4 Enzymes have active site, at which the substrate binds.

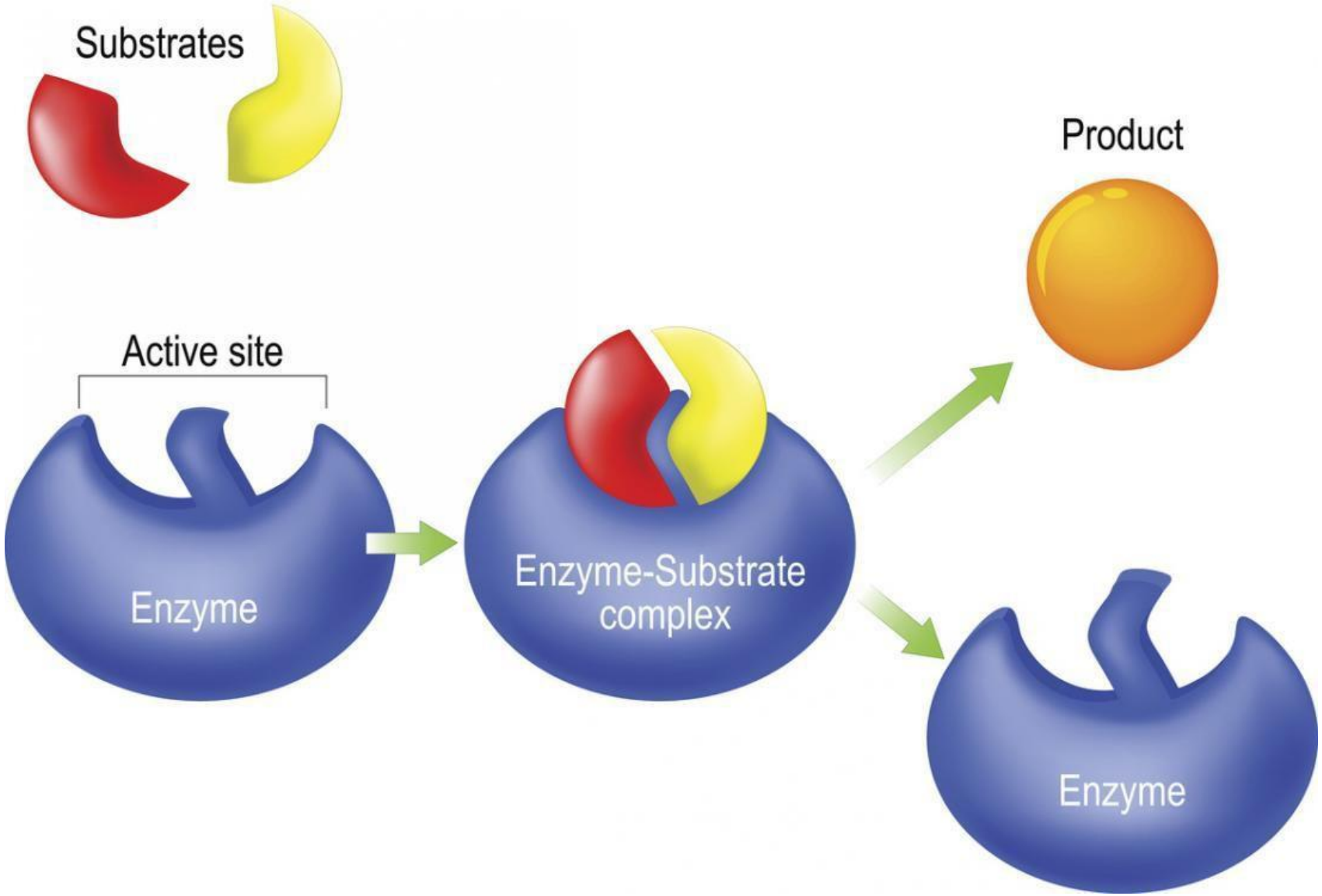
Important terms about enzymes

1 Active site : The place in the enzyme where the substrate binds.

- Enzyme molecules contain a special pocket called the active site.
- After substrate binds to the enzyme, it forms what is called enzyme-substrate complex (ES), and after that gives the product.

$$E + S \rightleftharpoons ES \rightarrow P$$

enzyme plus the substrate — enzyme-substrate complex — product


Substrates
Product
Active site
Enzyme
Enzyme-Substrate complex
Enzyme

Important terms about enzymes

2- Catalytic efficiency:

- Enzyme-catalyzed reactions are highly efficient, proceeding from 103–108 times faster than uncatalyzed reactions.
- This means that the presence of enzyme speeds the reaction.

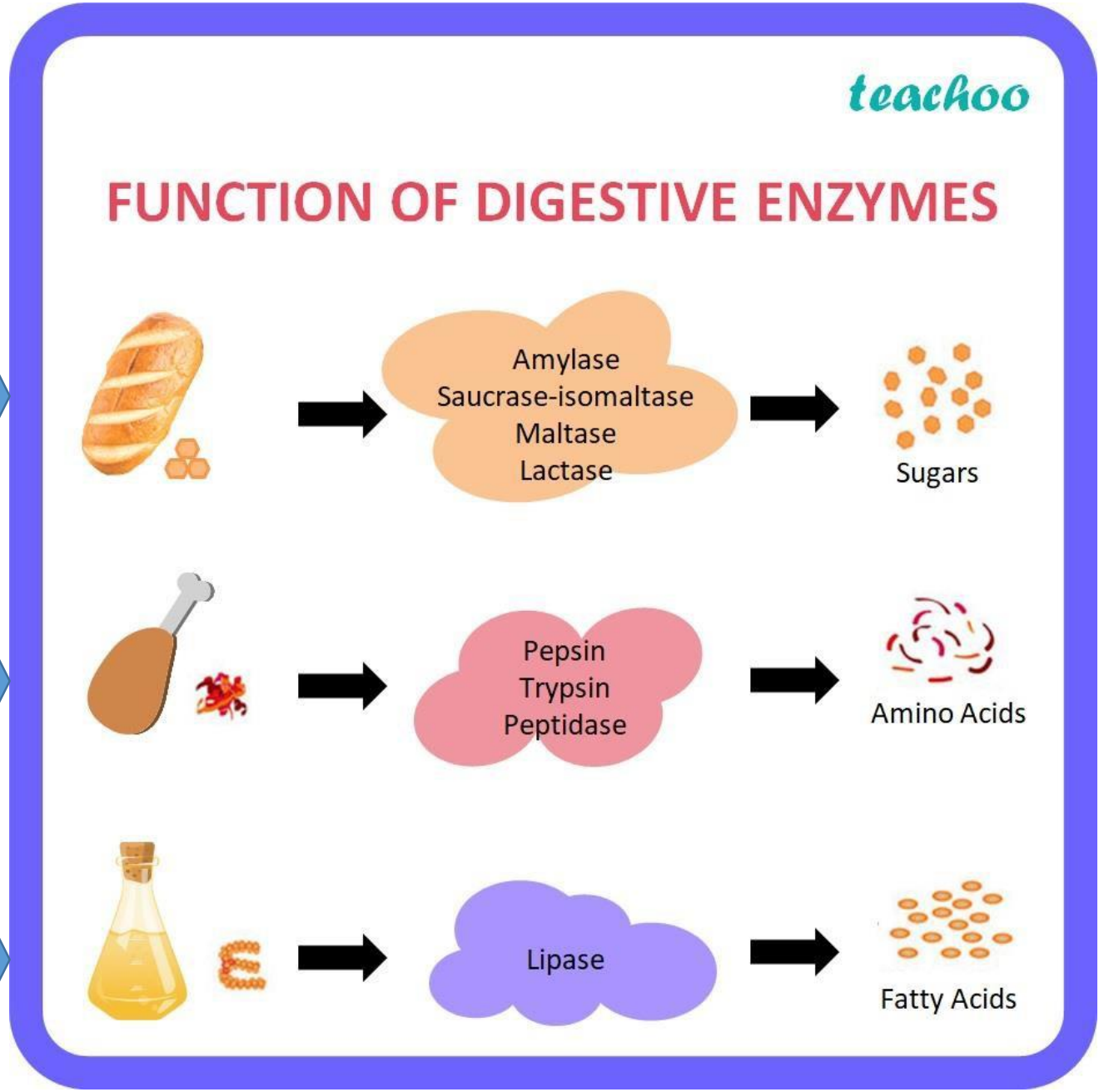
3-Enzyme specificity:

- Each enzyme has its substrate and its reaction.

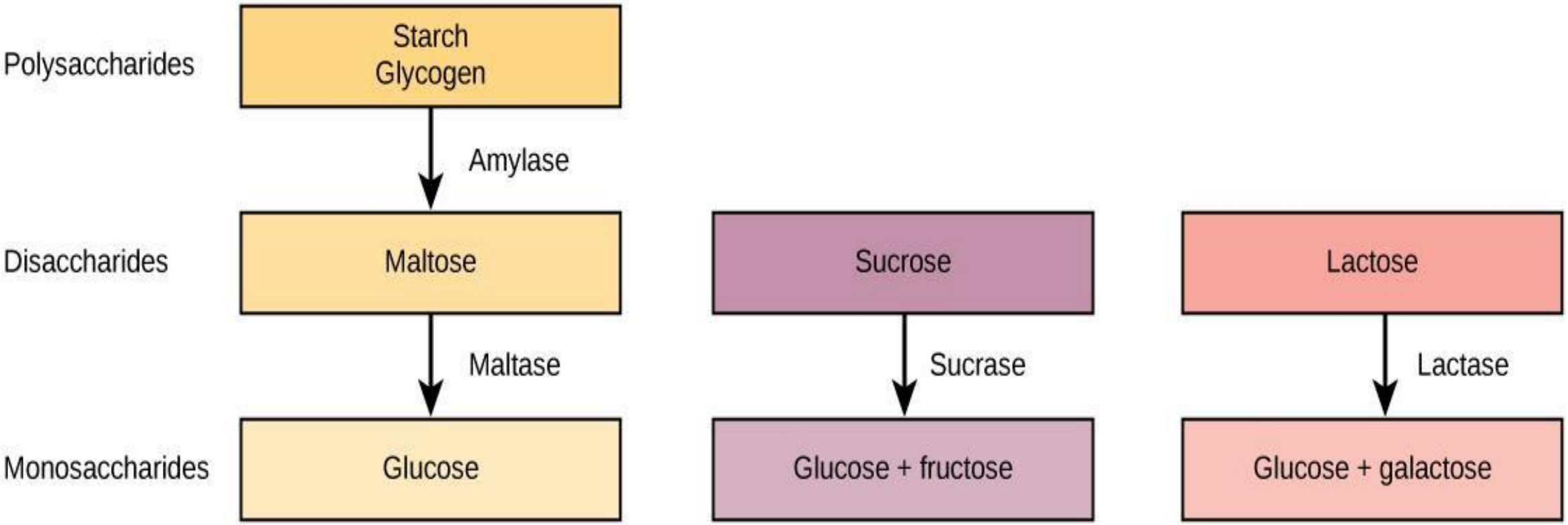
Carbohydrates

Proteins

Lipids



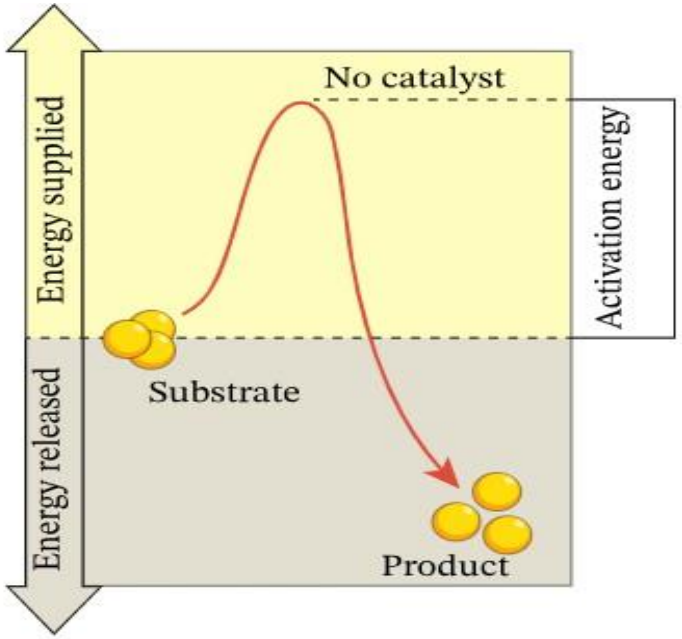
Enzymes that digest carbohydrates



Enzyme catalysis

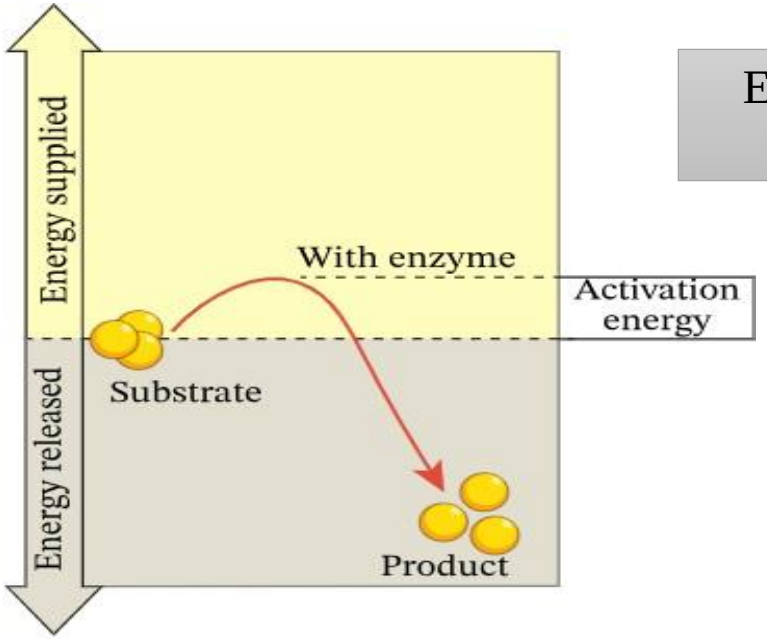
- Catalysis means the increasing of the rate or speed of the reaction.
- Enzyme catalysis is the process by which enzyme increases the reaction rate (accelerates reaction) without being changed.

Uncatalyzed reaction



Progress of reaction without enzyme

Enzyme catalyzed reaction



Progress of reaction with enzyme

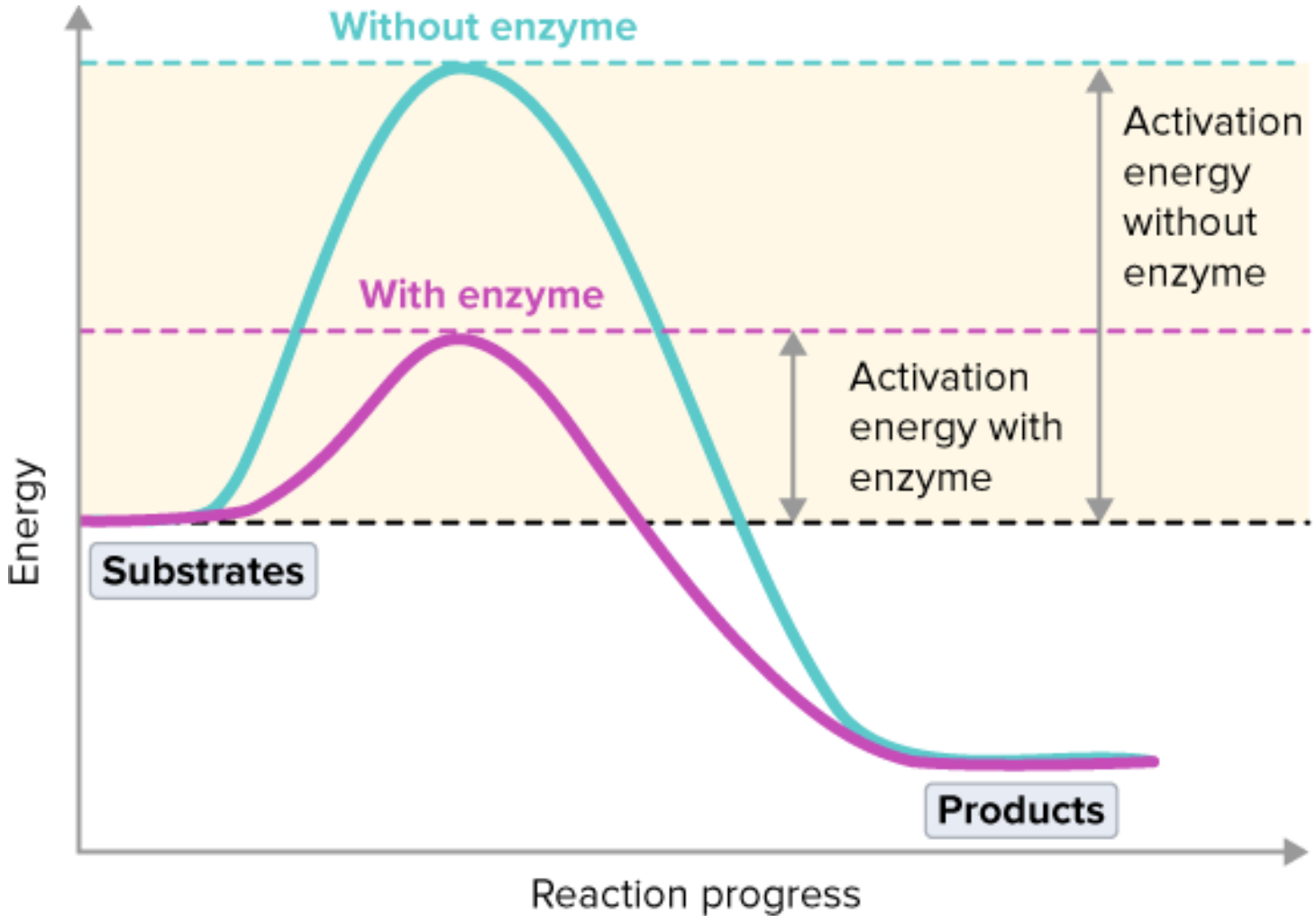
How enzyme catalyze the reaction

- **Enzyme is a catalyst that accelerates the rate of biochemical reaction by the decreasing activation energy.**

- **What is activation energy ?**

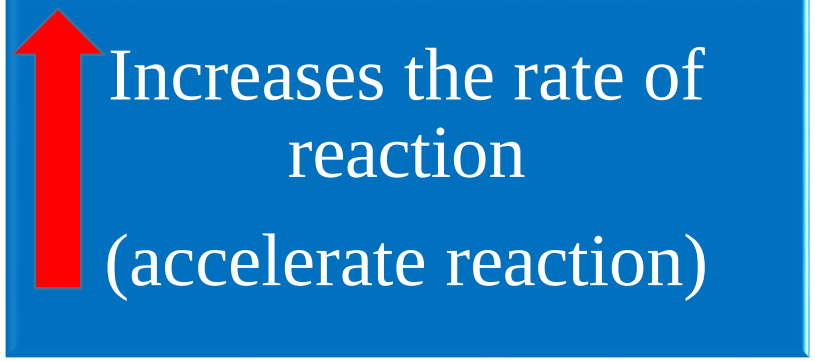
  - It is the amount of energy needed to from product.

- The role of enzyme is to reduce the amount of activation energy and speed the reaction.



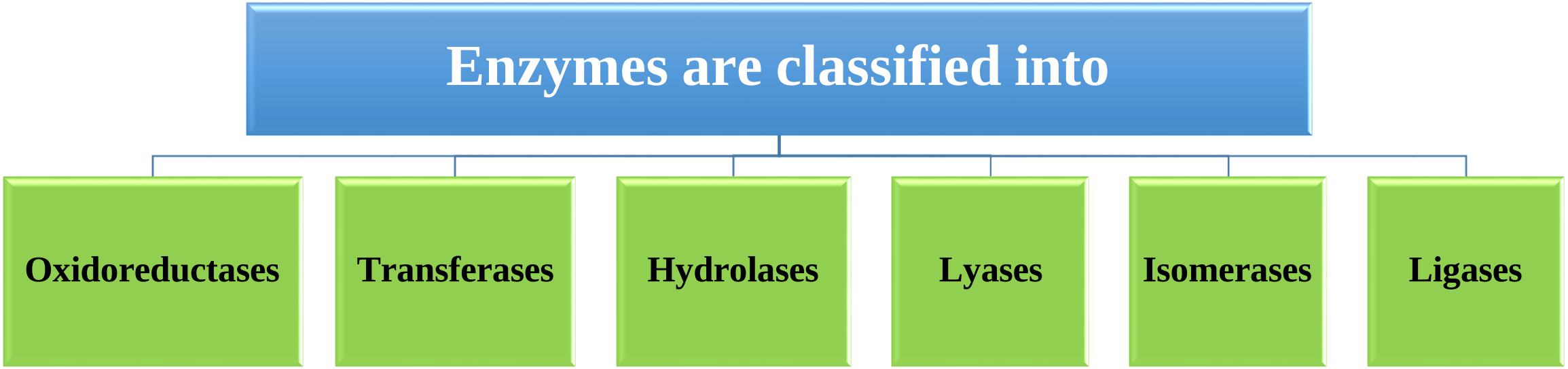
Enzyme decrease energy of activation, and increase the rate of reaction.

Enzyme


Increases the rate of reaction
(accelerate reaction)


Decrease
energy of activation

Classification of enzymes



Classification of enzymes

1- Oxidoreductases:

- One compound oxidized, and the other reduced (oxidation reduction reaction).

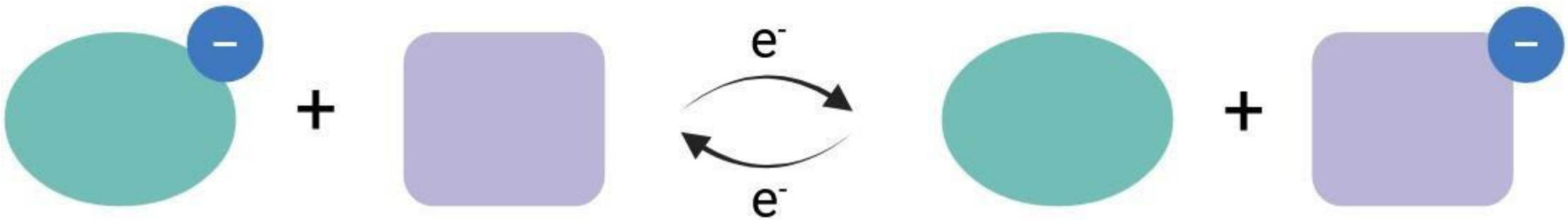
- Oxidoreductases catalyze the transfer of electrons from one molecule (called reductant or electron donor), to another, (called oxidant or called the electron acceptor).

$$A^- + B \rightarrow A + B^-$$

electron donor or reductant electron acceptor or oxidant

Oxidoreductase

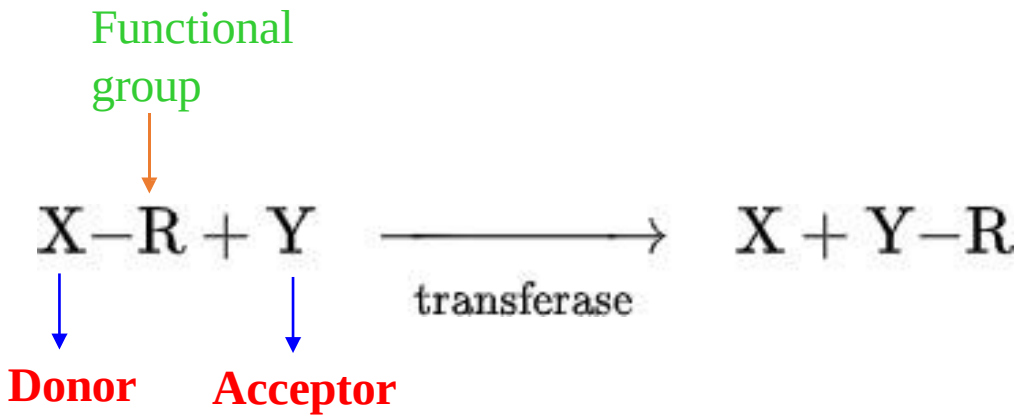
Catalyse electron transfer reactions (oxidation or reduction)



Classification of enzymes

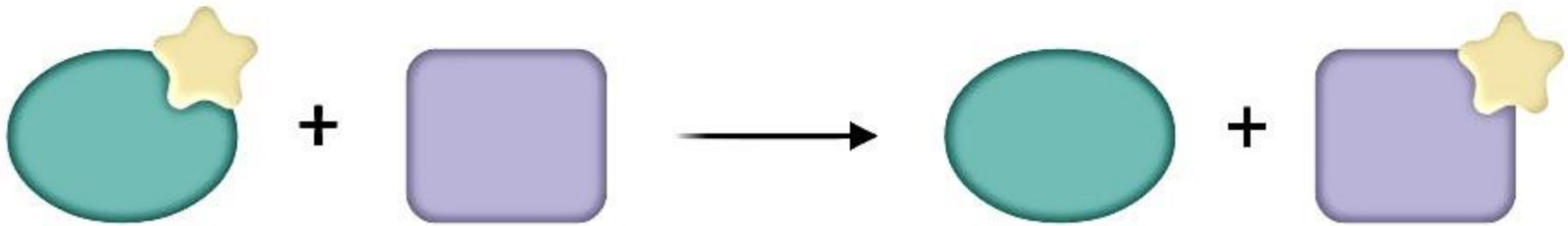
2- Transferases:

- Catalyze the transfer of specific functional group from one molecule (called the donor) to another (called the acceptor).



Transferase

Catalyse the transfer of specific functional groups from one donor to acceptor

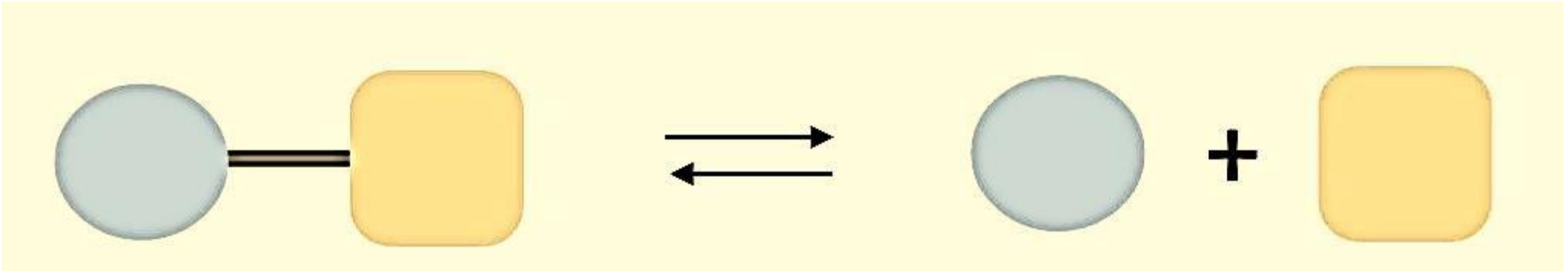


Classification of enzymes

3 Hydrolases: catalyze bond cleavages(break) by reaction with water (hydrolysis).

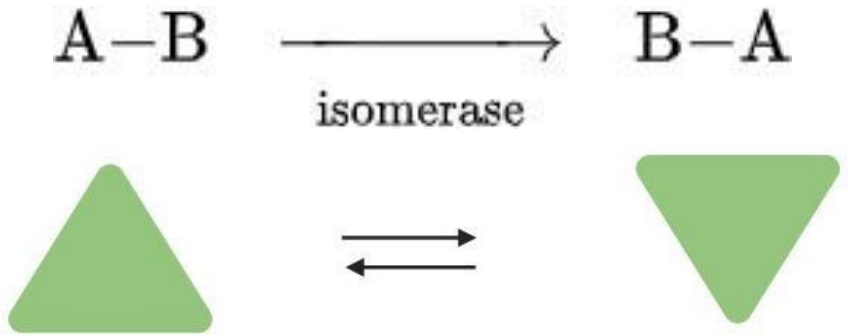
$$A{-}B + H_2O \xrightarrow[\text{hydrolase}]{} A{-}OH + B{-}H$$

4 Lyases: catalyze the breaking of chemical bonds (without the need of water) (lysis).

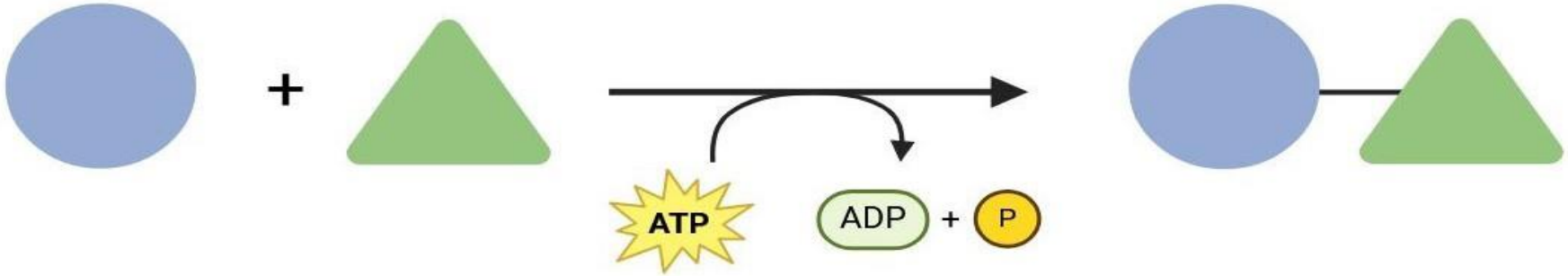


Classification of enzymes

5- Isomerases: catalyze the process of isomerization, which involves transforming a molecule into a different isomer.

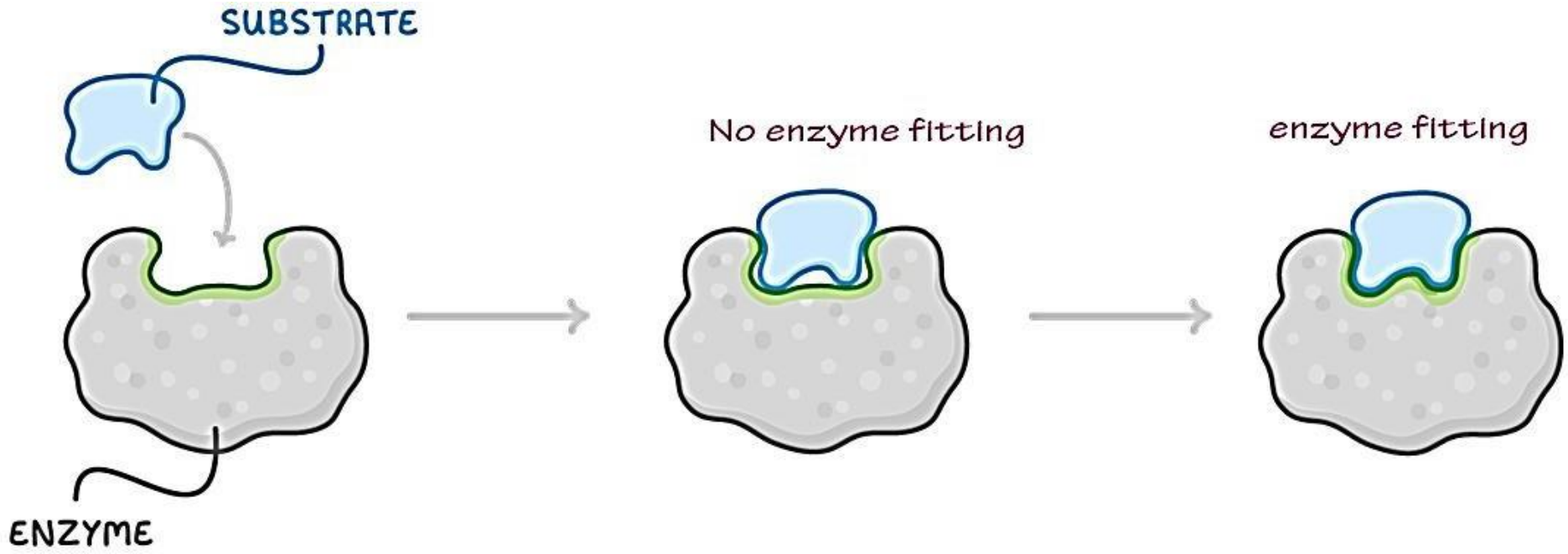


6- Ligases: catalyze the joining (**ligation**) of two molecules by **forming a new chemical bond**.



What is enzyme fitting ?

- When the substrate enters the active site of enzyme, it fits perfectly, and the two bind together, forming the enzyme-substrate complex.



Models of enzyme fitting

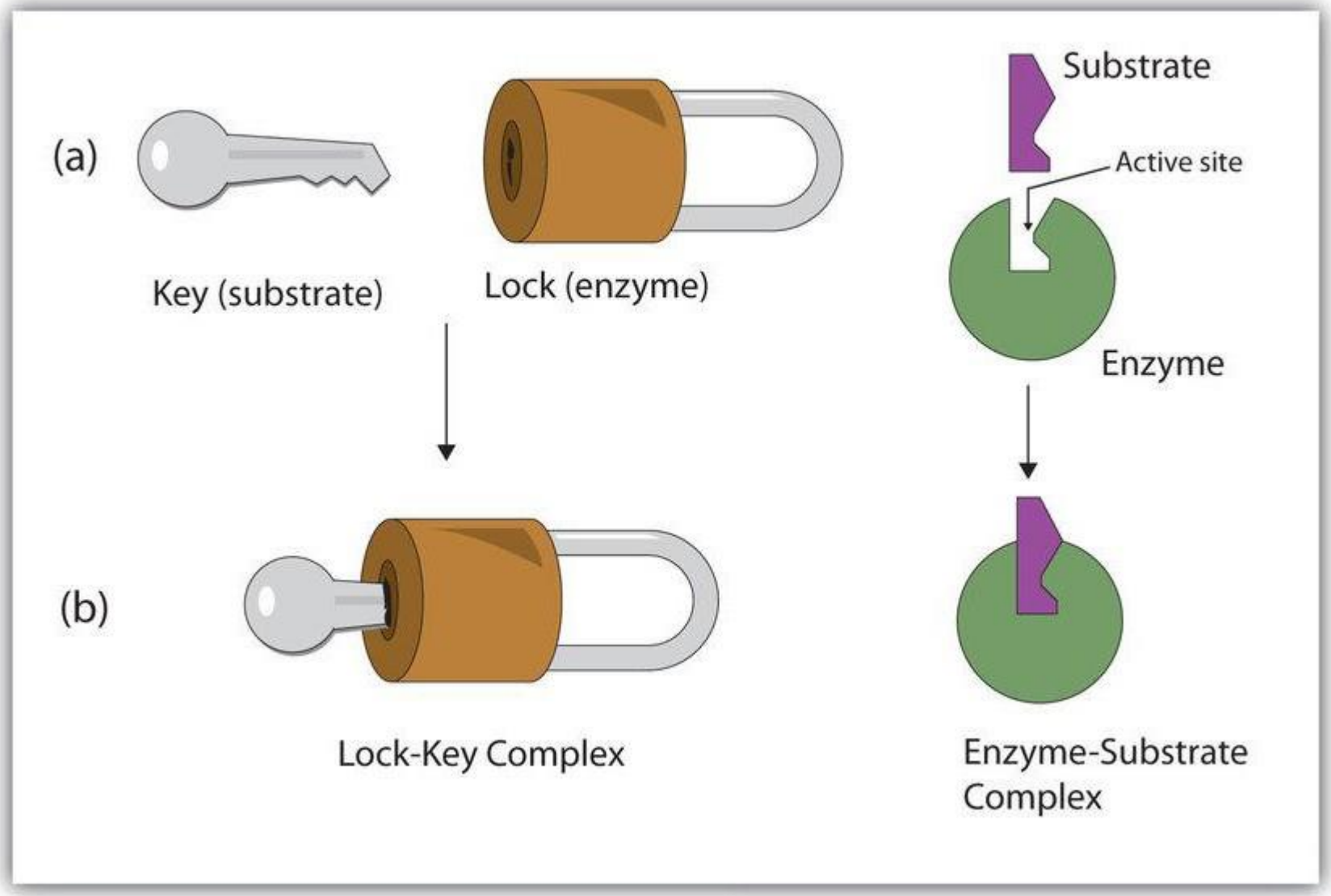
Lock and key model

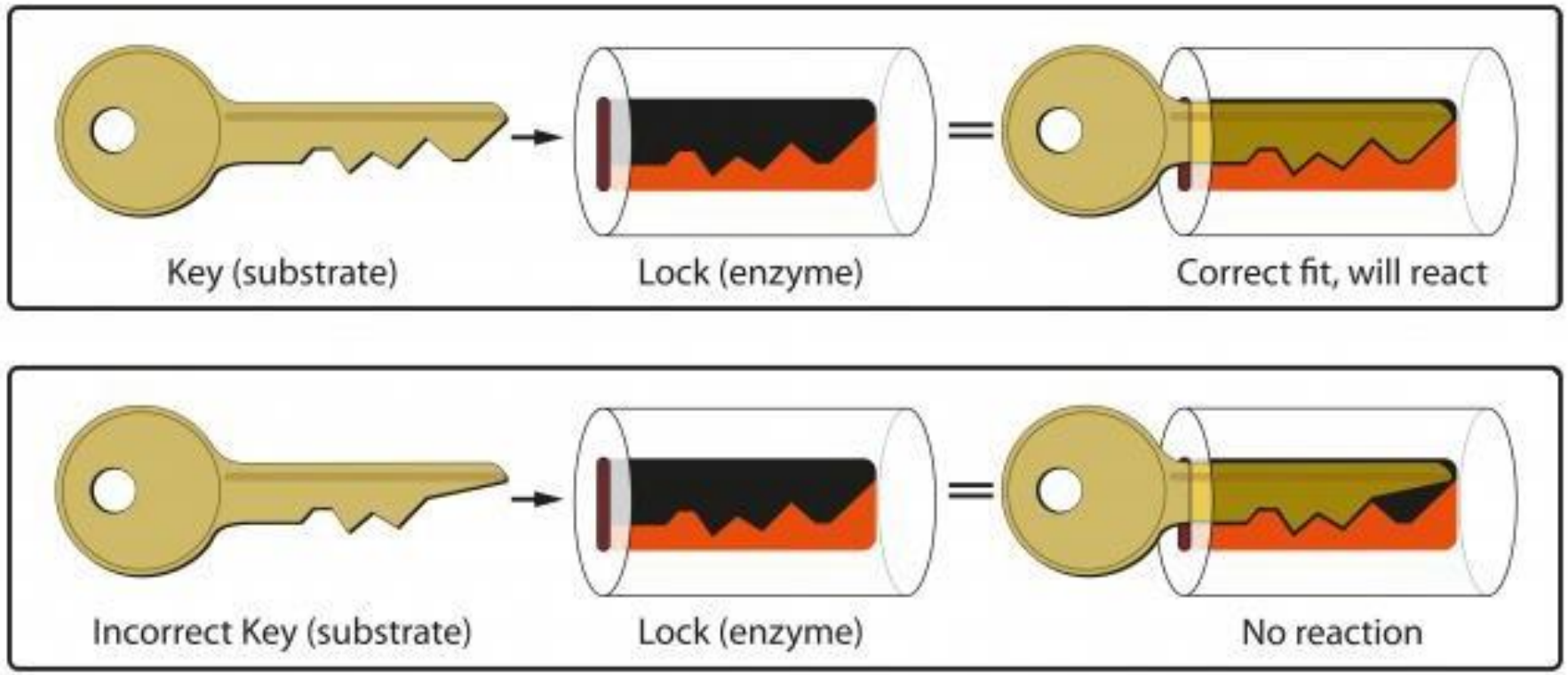


Lock and key model



- In this case, the enzyme is the lock, and the substrate is the key.
- Only the correct key (substrate) , enters the keyhole (the active site) of the lock (enzyme).
- The lock is? The keyhole is? The key is?
- Other keys may be too small, too large, or have wrong positioned teeth, so do not fit into the lock.
- Only the right-shaped and right-sized key can open the lock.


(a)
Key (substrate)
Lock (enzyme)
Substrate
Active site
Enzyme
(b)
Lock-Key Complex
Enzyme-Substrate Complex


Key (substrate)
Lock (enzyme)
Correct fit, will react
Incorrect Key (substrate)
Lock (enzyme)
No reaction

Questions

Complete the following statements:

1 In the lock and key model, …… is the lock, and …… is the key.

2 The enzyme that transfer specific functional group from one molecule to another is called……..

3 The substrate binds to enzyme at a specific location called ……..

4 The molecule that reacts with enzyme is called ……..

5 …… is the amount of energy needed to from or reach product.

6 ……enzyme catalyze the joining of two molecules by forming a new chemical bond.

Questions

Put (✓) or (X) for the following:

1 Lyase catalyzes bond cleavage by reaction with water. ( )

2 Enzymes are not specific to reactions or to substrates. ( )

3 Enzyme is needed in high amount or concentration in the reaction. ( )

4 Isomerases catalyze the joining of two molecules by forming a new chemical bond. ( )

5 Enzymes increase the rate of reaction and decrease the energy of activation. ( )

6 Enzyme is consumed or changed during the biochemical reaction. ( )


GOOD
LUCK